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A Universal Foundation Model for Zero-Shot Time Series Forecasting With Exogenous Variables

Fengqian Ding, Kechao Wang, Bo Li, Chen Xu, Yi Wu, and Hongchao Zhou

***Abstract*—Time series data play a critical role in industrial systems, enabling intelligent monitoring, optimization, and decision-making. Deep learning methods have achieved remarkable success in modeling complex, nonlinear relationships within domain-specific scenarios. However, their reliance on traditional training paradigms limits generalization across domains and constrains zero-shot forecasting capabilities. Although some foundation models for time series forecasting have emerged, they are primarily designed for univariate or general multivariate prediction tasks and do not explicitly model the asymmetric relationship between target and exogenous variables to support target forecasting. To this end, we propose UniCastX, a Transformer-based universal foundation model for time series forecasting with exogenous variables, pretrained on 26 billion time points from diverse real-world datasets using a tailored strategy. UniCastX employs attention mechanisms to capture temporal dependencies and leverage exogenous variables as auxiliary inputs, enhancing target prediction while reducing the influence of irrelevant or noisy factors. Extensive experiments on real-world benchmarks show that UniCastX achieves superior zero-shot forecasting performance over state-of-the-art foundation models and remains competitive with supervised baselines. It also offers low inference latency and strong robustness to exogenous noise, making it highly suitable for real-time industrial forecasting and decision-making applications.**

***Index Terms*—Exogenous variables, foundation model, time series forecasting, transformer, zero-shot forecast.**

I. Introduction

TIME series forecasting is a foundational challenge in data science, vital for areas, such as financial market forecasting, industrial production planning, and analysis of energy consumption [1], [2]. Recent studies [3], [4] have demonstrated that deep learning methods have markedly advanced this field, attributed to their superior scalability and capacity to model nonlinear relationships within domain-specific data, offering distinct advantages over classical models. Despite the considerable achievements of these methods in specific domains, they typically adhere to a traditional training mechanism, where they are trained to make predictions using the same datasets. Upon encountering new data, they require either retraining from scratch or fine-tuning. This approach limits their cross-domain generalizability and hinders their capability for universal zero-shot forecasting.

In recent years, the emergence of foundation models has revolutionized natural language processing (NLP) and computer vision [5], [6], largely due to their ability to leverage large-scale data to learn general-purpose representations transferable to diverse downstream tasks without additional training [7]. Inspired by this success, pretraining foundation models on large-scale datasets has attracted increasing interest in time series forecasting, enabling models to learn generalizable temporal representations reusable across diverse forecasting tasks and alleviating the traditional one-model-per-dataset paradigm. This paradigm is particularly appealing when task-specific data are scarce or constrained by privacy and copyright, and repeated retraining is costly. Although training a time series foundation model incurs higher upfront computational cost, this is often a one-time investment offset by strong generalization, enabling efficient reuse across tasks, lightweight fine-tuning, and forecasting in cold-start and data-scarce scenarios.

Motivated by these advantages, recent research has increasingly focused on developing pretrained foundation models specifically for time series forecasting by constructing architectures tailored to the intrinsic characteristics of time series and training them from scratch on large-scale datasets [8], [9]. In particular, concurrent work [10], [11], [12] has explored pretraining Transformer-based models designed for time series data on large corpora of real or synthetic time series, enabling more effective modeling of temporal dynamics and improved forecasting performance.

In real-world forecasting scenarios, systems are often only partially observed, making reliance solely on the target variable insufficient for accurate predictions. In many applications, multiple related variables are recorded, and the external covariates that influence the target but are not themselves the forecasting objective are referred to as *exogenous variables*. Their temporal realizations over time are denoted as *exogenous series*,

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which describe how these external factors evolve and provide additional contextual information for forecasting. For example, in industrial electricity demand forecasting, the future power consumption of a factory is influenced not only by its historical load but also by external factors, such as ambient temperature, production intensity, and operating schedules. These variables are not themselves the forecasting targets, yet their temporal variations provide important contextual information for anticipating changes in electricity demand. However, most existing foundation models for time series forecasting are designed for either univariate or general multivariate settings. Although multivariate foundation models can accommodate multiple input variables, they typically do not explicitly distinguish a designated forecasting target from auxiliary exogenous variables. In forecasting with exogenous variables, the target series remains the primary prediction objective, while the other variables serve as auxiliary information whose contributions should be selectively incorporated. Thus, forecasting with exogenous variables introduces additional challenges: although these external signals encode useful contextual dependencies, they may also introduce noise and uncertainty that hinder prediction quality. These issues become more pronounced for universal foundation models trained across diverse datasets, which must accommodate arbitrary numbers of exogenous variables and substantial cross-domain heterogeneity. Consequently, building a general model that can effectively harness exogenous information to improve target forecasting remains an open and challenging problem.

To this end, we propose UniCastX, a universal foundation model for time series forecasting with exogenous variables. Drawing inspiration from the successful application of Transformers in the time series domain, UniCastX is built upon the canonical Transformer encoder architecture with targeted modifications. Each encoder block includes channel-independent temporal self-attention and exogenous-to-target cross-attention. The temporal self-attention captures temporal dependencies within the target and exogenous series, while the cross-attention models the influence of exogenous series on the target series. By incorporating exogenous variables during model pretraining, UniCastX is able to learn generalizable patterns of how external driving factors influence temporal dynamics, supporting zero-shot forecasting on unseen datasets influenced by exogenous factors. In addition, Transformer-based networks are data-intensive and require extensive training data. Thus, we collected and processed time series data from multiple domains, totaling 26 billion time points, to train our model. During training, we balance the datasets through a resampling strategy to mitigate uneven data distribution and incorporate synthetic data to further enhance generalization. Overall, the contributions of this work are as follows.

1) We introduce UniCastX, a large-scale foundation model for time series forecasting with exogenous variables, trained on diverse real-world and synthetic datasets, totaling 26 billion time points. We have released our source code to facilitate further research and reproducibility.[1]
2) UniCastX is built on the canonical Transformer architecture, combining the proposed temporal self-attention and exogenous-to-target cross-attention to capture temporal dependencies and effectively exploit informative exogenous signals while mitigating interference from irrelevant or noisy variables.
3) Experimental results on seven real-world benchmarks show that UniCastX performs strongly against state-of-the-art (SOTA) foundation models in zero-shot forecasting and remains competitive with supervised baselines in in-distribution forecasting. Moreover, its low inference time and robustness to noisy exogenous variables demonstrate its practicality in real-world scenarios.

[1]https://github.com/suan-chang/rain-UniCastX.

II. Related Work

A. Deep Learning-Based Models for Time Series Forecasting

Deep learning-based models have achieved significant performance improvements across various time series forecasting tasks. Among these, RNN-based models, such as long short-term memory [13] and DeepAR [14], have been favored for their ability to capture temporal dependencies over time series. Convolutional neural network (CNN)-based models, like temporal convolutional network [15], leverage convolutional layers to capture local temporal patterns and improve forecasting performance. However, these models face limitations in capturing long-range dependencies, which hinders their performance in tasks that require modeling complex temporal patterns.

In particular, Transformer-based models leverage attention mechanisms to capture short- and long-term temporal dependencies, achieving strong performance in complex forecasting tasks. Informer [16], LogTrans [17], Pyraformer [18], and Autoformer [19] develop efficient attention modules to reduce the computational cost of long-sequence modeling. FEDformer [20] and TDformer [21] further exploit time- and frequency-domain features for forecasting. PatchTST [22] segments time series into patches to enlarge the receptive field and improve inference efficiency, while iTransformer [23] models serieswise representations to capture multivariate correlations. However, these models are primarily task-specific and require dataset-specific training, limiting their generalization to unseen datasets.

B. Foundation Models for Zero-Shot Forecasting

Despite the remarkable success of foundation models in NLP [24], their potential for zero-shot time series forecasting is still relatively unexplored. Nevertheless, there have been some notable recent efforts in this area. TimeGPT [25] introduced the first foundation model for time series, trained on diverse datasets and capable of making predictions on data not seen during training. ForecastPFN [12] introduced the first forecasting model trained purely on synthetic data, demonstrating the potential of synthetic data in building zero-shot forecasting models. TimesFM [11], a foundation model based on a decoder-only architecture, was trained on a large corpus of data from Google Trends, Wikipedia pageviews, and synthetic sources, enabling

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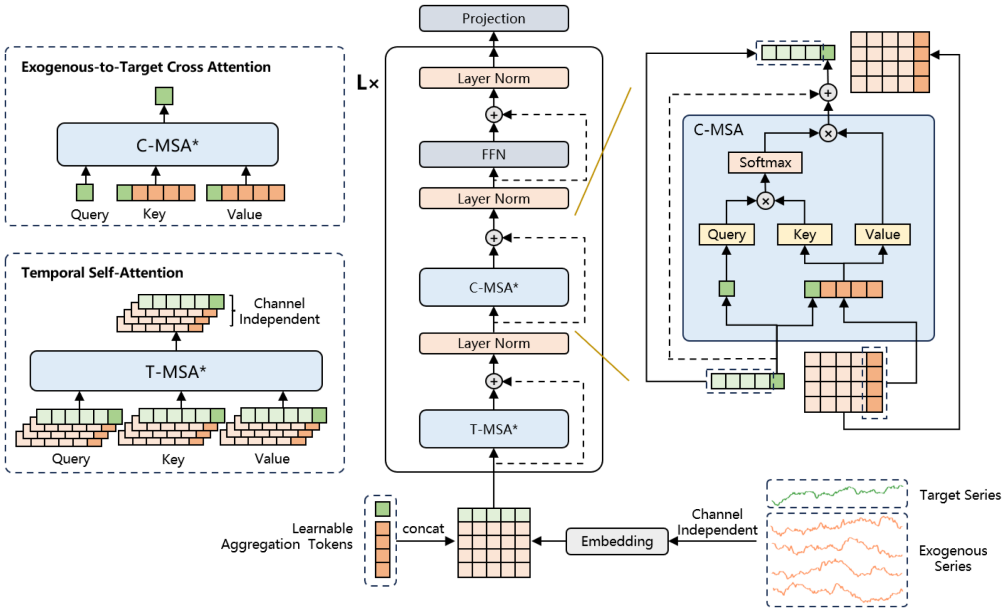


Fig. 1. UniCastX independently embeds the target and exogenous series, integrates them with aggregation tokens, and adopts an encoder-only architecture with shared temporal self-attention and exogenous-to-target cross-attention to model temporal and cross-variable dependencies. The aggregated target token is then projected to produce multistep forecasts.

autoregressive decoding for predictions. Lag-Llama [26] is a general-purpose foundation model for univariate probabilistic time series forecasting, utilizing a decoder-only Transformer architecture with lags as covariates. However, existing time series foundation models are primarily designed for univariate or general multivariate forecasting. Although multivariate models can accommodate multiple input series, they typically do not explicitly model the asymmetric relationship between a designated target and auxiliary exogenous variables. As a result, the role of exogenous information in supporting target forecasting remains underexplored.

III. Method

A. Problem Definition

The objective of this study is to develop a universal model for time series forecasting with exogenous series. Accordingly, given a target time series $\mathbf{x}_{1:L} = \{x_1, x_2, \ldots, x_L\} \in \mathbb{R}^{L\times 1}$ and a corresponding set of exogenous time series $\mathbf{e}_{1:L} = \{\mathbf{e}_{1:L}^{(1)}, \mathbf{e}_{1:L}^{(2)}, \ldots, \mathbf{e}_{1:L}^{(C)}\} \in \mathbb{R}^{L\times C}$, where L denotes the length of the look-back window, x_i indicates the value at the ith time point, C is the number of exogenous variables, $\mathbf{e}_{1:L}^{(j)}$ represents the jth exogenous series. The purpose of the forecasting model f is to predict future values over H time steps, represented as $\hat{\mathbf{x}}_{L+1:L+H} = \{\hat{x}_{L+1}, \hat{x}_{L+2}, \ldots, \hat{x}_{L+H}\}$, based on historical data. The formulation of the prediction task is as follows:

$$\hat{\mathbf{x}}_{L+1:L+H} = f(\mathbf{x}_{1:L}, \mathbf{e}_{1:L}). \tag{1}$$

B. Structure Overview

As shown in Fig. 1, UniCastX independently embeds target and exogenous series, integrates them with learnable aggregation tokens, and uses an encoder-only architecture with shared temporal and cross-attention layers to model their dependencies. The aggregated target token embedding is then projected to generate multistep forecasts. Fig. 2 further illustrates how the target and exogenous aggregation tokens summarize temporal information and facilitate exogenous-to-target interaction.

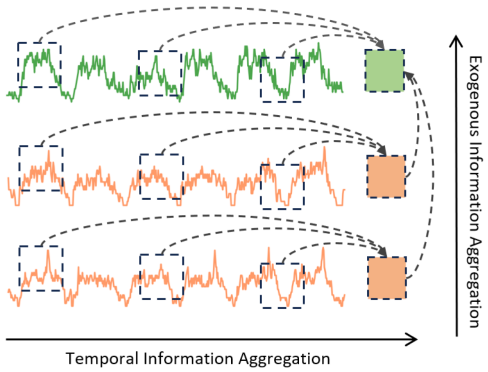


Fig. 2. Illustration of temporal and exogenous information aggregation. The aggregation token of each series captures its temporal information, while the target aggregation token further aggregates information from the exogenous series through cross-attention over the aggregation token embeddings.

C. Robust Normalization

Training a universal forecasting model requires handling time series from diverse domains and scales. Direct normalization or instance normalization [27] on windowed data are often ineffective, as heterogeneous outliers can distort scaling and degrade the preservation of essential data patterns. Therefore, we design a simple yet effective robust normalization approach. Specifically, during the training phase, each dataset is divided into multiple

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sliding windows. The mean $\mu(\mathbf{x}_{1:L})$ and standard deviation $\sigma(\mathbf{x}_{1:L})$ of the known historical portion of each window are computed. Subsequently, the entire window is renormalized using values within $\mu(\mathbf{x}_{1:L}) \pm K \times \sigma(\mathbf{x}_{1:L})$. Taking the target time series $\mathbf{x}_{1:L+H}$ as an example, the normalization of the window data is expressed as follows:

$$\mathbf{x}^{\text{norm}}_{1:L+H} = \frac{\mathbf{x}_{1:L+H} - \tilde{\mu}(\mathbf{x}_{1:L})}{\tilde{\sigma}(\mathbf{x}_{1:L}) + \epsilon} \tag{2}$$

where $\tilde{\mu}(\mathbf{x}_{1:L})$ and $\tilde{\sigma}(\mathbf{x}_{1:L})$ represent the mean and standard deviation computed after masking out values beyond the range $\mu(\mathbf{x}_{1:L}) \pm K \times \sigma(\mathbf{x}_{1:L})$, and ϵ is a small constant added for numerical stability. Subsequently, the historical portion $\mathbf{x}^{\text{norm}}_{1:L}$ is clipped to the range $[-K, K]$. For each sample, if any value in its $\mathbf{x}^{\text{norm}}_{L+1:L+H}$ portion falls outside the range, then the entire sample is masked out. The normalization process suppresses anomalies, stabilizes gradients, and ensures consistent scaling across datasets. For all exogenous time series $\mathbf{e}_{1:L+H}$, channelwise normalization is applied. During inference, we use reversible instance normalization [27] on inputs and denormalize predictions using input statistics, mitigating distribution shifts and supporting stable real-world deployment.

D. Input Embedding

Patch embedding: Following normalization, the time series is divided into nonoverlapping patches of length P, yielding $L_p = \lceil \frac{L-P}{S} \rceil + 1$ patches, where S is the stride. Prior studies [22], [28] show that patching aggregates local temporal information and accelerates inference by reducing the number of Transformer tokens. Both target and exogenous series are segmented into patches and then projected using a shared linear embedding, implemented via a channelwise 1-D convolution with kernel size P and stride S

$$\mathbf{E}_{\text{tar}}, \mathbf{E}_{\text{exo}} = \text{Conv1D}([\mathbf{x}; \mathbf{e}]) \tag{3}$$

where $\mathbf{E}_{\text{tar}} \in \mathbb{R}^{1 \times L_p \times D}$, $\mathbf{E}_{\text{exo}} \in \mathbb{R}^{C \times L_p \times D}$, D represents the hidden size of the embedding, and $[\cdot;\cdot]$ indicates concatenation operation along the channel dimension.

Aggregation token: Similarly to the class token used in Vision Transformers [29] and bidirectional encoder representations from transformers (BERT) [30] to aggregate global information, we append learnable aggregation tokens, $\mathbf{P}_{\text{tar}} \in \mathbb{R}^D$ and $\mathbf{P}_{\text{exo}} \in \mathbb{R}^D$, to the sequences of embedded patches of both the target and exogenous series, respectively. These aggregation tokens encapsulate the temporal information of each sequence, serving as global representations for the target and exogenous series. Appending these tokens results in the embeddings $\hat{\mathbf{E}}_{\text{tar}} = [\mathbf{E}_{\text{tar}}; \mathbf{P}_{\text{tar}}] \in \mathbb{R}^{1 \times (L_p+1) \times D}$ and $\hat{\mathbf{E}}_{\text{exo}} = [\mathbf{E}_{\text{exo}}; \mathbf{P}_{\text{exo}}] \in \mathbb{R}^{C \times (L_p+1) \times D}$, which are used to model the exogenous-to-target interactions for subsequent processing.

All exogenous variables share the same learnable aggregation token P_{exo}, which is appended to each exogenous sequence and yields a series-specific contextualized representation after temporal attention. This design enables flexible interaction modeling between the target and exogenous variables, supporting an arbitrary number of exogenous inputs during inference. We set a maximum of eight exogenous variables during training. If the number exceeds this limit, then excess variables are randomly selected, while missing ones are zero-padded. In addition, timestamp encodings, such as second of day, day of week, and month, are treated as exogenous variables for all datasets and transformed into six features using sine and cosine functions [10].

E. Encoder-Only Architecture

We employ an encoder-only architecture for direct multistep forecasting with exogenous variables, thereby avoiding iterative autoregressive decoding [23], [31]. In iterated multistep forecasting, prediction errors are recursively propagated to subsequent steps and can accumulate as the forecasting horizon increases, whereas direct multistep forecasting avoids such iterative error propagation [32]. Each encoder layer primarily includes a temporal attention module, a cross-attention module, and a feedforward network. The temporal attention is used to model the time dependencies, while the cross-attention focuses on the influence of exogenous variables on the target variable.

Encoder layer: In the temporal attention module, $\hat{\mathbf{E}}_{\text{tar}}$ and $\hat{\mathbf{E}}_{\text{exo}}$ are concatenated along the channel dimension to form $[\hat{\mathbf{E}}^0_{\text{tar}}; \hat{\mathbf{E}}^0_{\text{exo}}] \in \mathbb{R}^{(1+C) \times (L_p+1) \times D}$, serving as input. Specifically, we apply multihead self-attention (MSA) to capture the temporal dependencies for each channel independently. For a batch of size B, the input shape of $B \times (1+C) \times (L_p+1) \times D$ is first reshaped to $(B \cdot (1+C)) \times (L_p+1) \times D$ to enable independent channel modeling. After temporal attention, the output is reshaped back to $B \times (1+C) \times (L_p+1) \times D$. This approach of maintaining channel independence ensures that the modeling of each channel is not influenced by others, preventing channel-mixing that could adversely affect the modeling of the target series. The operations within the temporal attention are structured as follows:

$$\tilde{\mathbf{E}}^n_{\text{tar}}, \tilde{\mathbf{E}}^n_{\text{exo}} = \text{LN}\left(\text{T-MSA}\left(\left[\hat{\mathbf{E}}^{n-1}_{\text{tar}}; \hat{\mathbf{E}}^{n-1}_{\text{exo}}\right]\right) + \left[\hat{\mathbf{E}}^{n-1}_{\text{tar}}; \hat{\mathbf{E}}^{n-1}_{\text{exo}}\right]\right) \tag{4}$$

where $n \in \{1, 2, \ldots, N\}$ represents the nth encoder layer, and N is the number of encoder layers. LN denotes layer normalization, and here we utilize postnormalization for stable training. Temporal multihead self-attention (T-MSA) refers to the temporal attention mechanism, omitting the index for each head, and can be expressed as follows:

$$\text{T-MSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathcal{R}(\mathbf{Q})\,\mathcal{R}(\mathbf{K})^T}{\sqrt{D}} + \mathbf{M}_t\right)\mathbf{V} \tag{5}$$

where $\mathcal{R}(\cdot)$ denotes the rotation transformation used in the rotary positional embedding [33], which helps preserve relative positions in temporal dependencies while also improving the model's extrapolation capabilities. $\mathbf{M}_t \in \mathbb{R}^{(L_p+1) \times (L_p+1)}$ represents the temporal mask matrix, which is used during training to randomly mask earlier patches in the sequence, introducing variation in the input length and enhancing the model's ability to adapt to different context lengths. Specifically, during training, earlier patches are masked with a probability of $\alpha = 0.25$, and the maximum number of masked patches is set to half of L_p.

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As the primary focus is on the target variable, we incorporate exogenous variables as auxiliary information via cross-attention, following [31]. Unlike [31], which is limited to task-specific settings and fixed input lengths, our design supports arbitrary-length inputs and allows exogenous variables to participate in temporal attention, enabling better modeling of dynamic dependencies. Specifically, the aggregation token embeddings for the target and exogenous series encapsulate the information of their respective sequences after temporal attention. To further extract relevant information for predicting the target series, these embeddings $\tilde{\mathbf{P}}^n_{\text{tar}}$ and $\tilde{\mathbf{P}}^n_{\text{exo}}$ are retrieved from $\tilde{\mathbf{E}}^n_{\text{tar}}$ and $\tilde{\mathbf{E}}^n_{\text{exo}}$ and processed via cross-attention, as detailed in the following process:

$$\hat{\mathbf{P}}^n_{\text{tar}} = \text{LN}\left(\text{C-MSA}\left(\tilde{\mathbf{P}}^n_{\text{tar}}, \left[\tilde{\mathbf{P}}^n_{\text{tar}}; \tilde{\mathbf{P}}^n_{\text{exo}}\right], \left[\tilde{\mathbf{P}}^n_{\text{tar}}; \tilde{\mathbf{P}}^n_{\text{exo}}\right]\right) + \tilde{\mathbf{P}}^n_{\text{tar}}\right) \tag{6}$$

where C-MSA, omitting the index for each head, can be expressed as

$$\text{C-MSA}\left(\mathbf{Q}, \mathbf{K}, \mathbf{V}\right) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D}} + \mathbf{M}_c\right)\mathbf{V}. \tag{7}$$

In this case, $\tilde{\mathbf{P}}^n_{\text{tar}}$ serves as the query, while $[\tilde{\mathbf{P}}^n_{\text{tar}}; \tilde{\mathbf{P}}^n_{\text{exo}}]$ acts as both the key and value. Here, $\mathbf{M}_c \in \mathbb{R}^{1\times(C+1)}$ is a mask that sets the attention scores to $-\infty$ for padded exogenous variables, ensuring that they do not contribute to the softmax output in the training phase. The utilization of aggregation token embeddings in cross-attention ensures that the influence of exogenous variables on the target is permutation invariant. This means that the contribution of exogenous variables does not depend on the order of the input data, thus maintaining the consistency of their impact regardless of the indexing of the exogenous variables. Since cross-attention uses a single target aggregation token as the query and $C+1$ aggregation tokens as keys and values, its computational overhead scales linearly with the number of exogenous variables C for a fixed embedding dimension. In addition, the temporal and cross-attention mechanisms share the same weights but are applied to different input dimensions. Empirically, this operation improves the model's ability to perform temporal modeling while maintaining parameter efficiency [34].

The output $\hat{\mathbf{P}}^n_{\text{tar}}$ updates the aggregation token embedding in $\tilde{\mathbf{E}}^n_{\text{tar}}$, which, along with $\tilde{\mathbf{E}}^n_{\text{exo}}$, serves as the input to the feedforward layer. This is formalized as

$$\hat{\mathbf{E}}^n_{\text{tar}}, \hat{\mathbf{E}}^n_{\text{exo}} = \text{LN}\left(\text{FFN}\left(\left[\tilde{\mathbf{E}}^n_{\text{tar}}; \tilde{\mathbf{E}}^n_{\text{exo}}\right]\right) + \left[\tilde{\mathbf{E}}^n_{\text{tar}}; \tilde{\mathbf{E}}^n_{\text{exo}}\right]\right). \tag{8}$$

The output $\hat{\mathbf{E}}^n_{\text{tar}}$ and $\hat{\mathbf{E}}^n_{\text{exo}}$ are used as the input for the next encoder layer. Overall, the process within an encoder layer can be simplified as

$$\hat{\mathbf{E}}^n_{\text{tar}}, \hat{\mathbf{E}}^n_{\text{exo}} = \text{EncoderLayer}(\hat{\mathbf{E}}^{n-1}_{\text{tar}}, \hat{\mathbf{E}}^{n-1}_{\text{exo}}), \quad n \in 1, 2, \ldots, N. \tag{9}$$

Forecasting: After passing through multiple encoder layers, the aggregation token embedding of target series establishes temporal dependencies and adaptively aggregates information from the exogenous series. Consequently, the output from the last encoder layer, $\hat{\mathbf{P}}^N_{\text{tar}}$, is passed through a linear layer to predict the results of the target series for the subsequent H time steps. The procedure is formalized as follows:

$$\hat{\mathbf{x}} = \text{Projection}\left(\hat{\mathbf{P}}^N_{\text{tar}}\right). \tag{10}$$

F. Training

When training a universal forecasting model with extensive cross-domain data, a straightforward approach is to sequentially feed data from each dataset to the model in each epoch. However, this method can lead to unstable learning and catastrophic forgetting [35]. To address these issues and enhance the model's generalization capabilities, we propose a simple yet effective probabilistic resampling strategy to balance the influence of datasets with varying sizes. Specifically, for a given set of datasets $\mathcal{D} = \{D_1, D_2, \ldots, D_n\}$, where each dataset D_i contains W_i data windows, the initial probability of selecting dataset D_i is defined as $P(D_i) = \frac{W_i}{\sum_{j=1}^n W_j}$. To further address disparities in dataset sizes, we apply a root transformation followed by rescaling: $P'(D_i) = \frac{P(D_i)^{1/\gamma}}{\sum_{j=1}^n P(D_j)^{1/\gamma}}$, where γ is the root transformation parameter and is set to 2 for square root normalization. Under this formulation, the probability of selecting a specific data window w^i_k from dataset D_i is given by $P(w^i_k) = P'(D_i) \cdot \frac{1}{W_i}$. This resampling strategy effectively reduces the overrepresentation of datasets with a large number of windows, ensuring a more balanced contribution from all datasets during training.

In addition, we employ the mean absolute error (MAE) as the training loss function and the loss function is defined as follows, since MAE offers better robustness to heavy-tailed noise and stabilizes optimization across heterogeneous datasets

$$\mathcal{L} = \frac{1}{H}\sum_{i=1}^{H}|\hat{x}_{L+i} - x_{L+i}| \tag{11}$$

where $\hat{x}_{L+i}$ represents the predicted value from the model for the future time step, and x_{L+i} is the actual observed value at the time step.

IV. Experiment

A. Experimental Settings

Datasets: Training a general time series forecasting model requires data from diverse sources to ensure robust performance across various forecasting tasks [25]. To this end, we have collected and processed time series from multiple domains, totaling 26B observations. These data are sourced from publicly available real-world repositories across a wide range of application domains and collectively contain 69 datasets, which are used exclusively for pretraining the foundation model. The statistical summary of these datasets is presented in Table I, where the data are summarized at the domain level. For each individual dataset, we use 90% of the time points for training and the remaining 10% for validation. Moreover, prior work has shown that incorporating synthetic data can further increase dataset diversity and improve generalization [12], [36]. Consequently, we follow [12] to generate synthetic data at three common real-world frequencies: daily, hourly, and minutely. In addition to the pretraining corpus, we reserve seven widely used benchmark

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TABLE I
STATISTICS OF COLLECTED TIME SERIES DATA BY DOMAIN

Domain	Observations	Percentage (%)
Energy	14 164 830 248	53.30
Climate	6 111 604 307	22.99
Web	4 316 130 725	16.24
Transportation	1 860 793 437	7.00
Synthetic Data	58 560 000	0.22
Electric Power	49 156 633	0.19
Environment	6 922 482	0.03
Finance	4 832 045	0.02
Industry	1 516 948	0.01

TABLE II
COMPARISON BETWEEN SOTA FOUNDATION MODELS

Model	Params	Arch.	Data	Exog.
Chronos	710M	Enc-Dec	84B	✗
Moirai	311M	Enc	27B	✓
GTT	57M	Enc	200M	✓
TimesFM	200M	Dec	100B	✗
ForecastPFN	–	Enc	60M	✗
UniCastX	56M	Enc	26B	✓

datasets for zero-shot evaluation. These benchmark datasets are strictly held out from the pretraining corpus to ensure a valid zero-shot setting. They include the four electricity transformer temperature (ETT) datasets as well as the Electricity, Traffic, and Weather datasets [16], [19]. These datasets are split into training, validation, and test sets in a 7:1:2 ratio.

Baselines: To evaluate the performance of our model, we have applied 13 different comparisons. For zero-shot evaluation, we select five SOTA pretrained foundation models, including Moirai [37], Chronos [36], TimesFM [11], general time transformer (GTT) [10], and ForecastPFN [12]. The comparison between UniCastX and representative SOTA pretrained foundation models is summarized in Table II, which reports their architectural design, pretraining data scale, maximum model size, and whether exogenous variables are explicitly supported as inputs. For in-distribution forecasting evaluation, the models used include eight types that are Transformer-based, Linear-based, and CNN-based, namely, TimeXer [31], iTransformer [23], TimesNet [38], LightTS [39], DLinear [32], PatchTST [22], Autoformer [19], and Informer [16].

Implementation details: During the pretraining process, the input length is fixed at 512, and the forecast horizon is set at 128. We apply a learning rate scheduler that includes a linear warmup for the first 5% of the steps, followed by cosine annealing. The AdamW optimizer is used with a learning rate of 1e-4, weight decay of 1e-3, and $\beta_1 = 0.9$ and $\beta_2 = 0.99$. The batch size is set to 1024. For fine-tuning, we use the Adam optimizer with a learning rate of 1e-5 and a batch size of 64, stopping when the validation loss increases. The model is pretrained on 8 Nvidia A800 GPUs using bfloat16 precision. The version of the model that achieves the best validation loss is selected for subsequent experiments.

B. Performance on Zero-Shot Forecasting

In this section, we compare the performance of UniCastX with SOTA pretrained foundation models in zero-shot forecasting scenario. Supervised baselines are excluded because they require task-specific training on the benchmark datasets, which is incompatible with the zero-shot setting. Most of the comparative pretrained foundation models, including Chronos, TimesFM, and ForecastPFN, do not explicitly support exogenous variables as inputs. To ensure a fair and controlled comparison, we therefore conduct the evaluation under a univariate forecasting setting, where the "OT" feature from each benchmark dataset is used as the prediction target. All models are set with an input length of 512 to predict various horizons. Since all seven benchmark datasets are strictly held out from pretraining, this zero-shot evaluation also represents a dataset-level cold-start setting, where UniCastX is directly applied to unseen target datasets without task-specific training or parameter updates. The results for different forecasting horizons are shown in Table III. Although UniCastX is designed to leverage information from exogenous variables to enhance the predictive performance of the target series, it still achieves the best results on most datasets compared to models with larger numbers of parameters and training samples. Specifically, our method achieves the best performance in 20 cases across all experiments, while the second-best, $\text{Chronos}_{\text{large}}$, only achieves the best performance in nine cases. However, $\text{Chronos}_{\text{large}}$ has $12\times$ the parameters and $3\times$ the training data volume of our model and does not support exogenous variable input. By independently embedding the target and exogenous sequences and applying temporal attention, our model can accommodate inputs with an arbitrary number of variables. In addition, when applied to univariate input, the cross-attention mechanism becomes an additional linear transformation of the target sequence, further enhancing prediction performance.

Furthermore, to validate UniCastX's performance with exogenous variables, we conduct additional comparisons with Moirai and GTT, both of which support multivariate inputs. The results of these comparative experiments are presented in Table IV. From the results in Tables III and IV, it is evident that incorporating exogenous variables further improves the performance of our model. This enhancement validates that our model architecture effectively extracts useful information from exogenous variables to improve the prediction of the target variable. In contrast, Moirai and GTT treat both target and exogenous variables equally, which can introduce unnecessary information or noise into the prediction of the target variable. Therefore, the proposed architecture, together with the training strategy, enables UniCastX to function as a general-purpose foundation model with flexible input configurations, achieving robust and competitive performance across both univariate and exogenous-variable forecasting scenarios.

C. Performance on In-Distribution Forecasting

In this section, we compare UniCastX against supervised forecasting models on in-distribution forecasting tasks. All supervised methods are trained on the training split of each dataset for various forecasting horizons. To ensure consistency, all comparative methods adopt a multivariate input, univariate output format for forecasting with exogenous variables. For our model, we report results for both zero-shot and fine-tuned scenarios.

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TABLE III
EXPERIMENTAL RESULTS OF ZERO-SHOT FORECASTING WITHOUT EXOGENOUS VARIABLES

Dataset	Models	UniCastX		$\text{Chronos}_{\text{small}}$		$\text{Chronos}_{\text{base}}$		$\text{Chronos}_{\text{large}}$		$\text{Moirai}_{\text{small}}$		$\text{Moirai}_{\text{base}}$		$\text{Moirai}_{\text{large}}$		GTT_{tiny}		$\text{GTT}_{\text{small}}$		$\text{GTT}_{\text{large}}$		TimesFM		ForecastPFN	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	64	0.083	0.219	0.089	0.225	0.088	0.221	0.085	0.217	0.091	0.228	0.087	0.223	0.087	0.223	0.101	0.240	0.091	0.230	0.096	0.236	0.104	0.244	0.208	0.363
	96	0.105	0.248	0.109	0.252	0.109	0.249	0.105	0.244	0.113	0.257	0.111	0.252	0.113	0.255	0.121	0.267	0.114	0.260	0.114	0.260	0.138	0.283	0.206	0.360
	128	0.117	0.265	0.121	0.268	0.119	0.261	0.116	0.258	0.128	0.276	0.117	0.268	0.124	0.269	0.131	0.279	0.128	0.276	0.120	0.270	0.168	0.311	0.197	0.356
	Avg	0.102	0.244	0.106	0.248	0.105	0.244	0.102	0.240	0.111	0.254	0.105	0.248	0.108	0.249	0.118	0.262	0.111	0.255	0.110	0.255	0.137	0.279	0.204	0.360
ETTh2	64	0.195	0.328	0.236	0.363	0.212	0.341	0.199	0.333	0.229	0.362	0.232	0.359	0.226	0.356	0.234	0.360	0.225	0.356	0.223	0.352	0.225	0.345	0.613	0.625
	96	0.237	0.368	0.291	0.409	0.261	0.386	0.250	0.379	0.273	0.399	0.278	0.398	0.285	0.402	0.280	0.400	0.276	0.400	0.261	0.386	0.280	0.388	0.604	0.622
	128	0.259	0.389	0.322	0.436	0.287	0.410	0.280	0.406	0.306	0.425	0.297	0.418	0.311	0.426	0.306	0.423	0.308	0.429	0.276	0.403	0.309	0.412	0.587	0.613
	Avg	0.230	0.362	0.283	0.403	0.253	0.379	0.243	0.373	0.269	0.395	0.269	0.392	0.274	0.395	0.273	0.394	0.270	0.395	0.253	0.380	0.271	0.382	0.601	0.620
ETTm1	64	0.037	0.138	0.039	0.141	0.037	0.136	0.038	0.138	0.045	0.153	0.045	0.152	0.044	0.149	0.049	0.159	0.041	0.144	0.044	0.153	0.044	0.149	0.171	0.316
	96	0.048	0.159	0.050	0.163	0.046	0.155	0.047	0.159	0.054	0.169	0.053	0.169	0.053	0.166	0.061	0.181	0.055	0.169	0.054	0.174	0.056	0.170	0.171	0.318
	128	0.058	0.176	0.060	0.180	0.055	0.172	0.056	0.175	0.067	0.190	0.065	0.188	0.067	0.188	0.074	0.201	0.068	0.190	0.064	0.191	0.068	0.189	0.201	0.354
	Avg	0.048	0.158	0.050	0.161	0.046	0.154	0.047	0.157	0.055	0.171	0.054	0.170	0.055	0.168	0.061	0.180	0.055	0.168	0.054	0.173	0.056	0.169	0.181	0.329
ETTm2	64	0.102	0.221	0.187	0.288	0.182	0.283	0.154	0.263	0.234	0.342	0.236	0.339	0.215	0.322	0.131	0.248	0.117	0.236	0.120	0.239	0.144	0.262	0.568	0.600
	96	0.125	0.251	0.222	0.329	0.217	0.326	0.186	0.303	0.208	0.329	0.225	0.341	0.221	0.337	0.153	0.275	0.153	0.275	0.145	0.273	0.167	0.288	0.588	0.614
	128	0.148	0.277	0.244	0.352	0.240	0.350	0.207	0.326	0.246	0.362	0.255	0.368	0.246	0.362	0.183	0.306	0.184	0.306	0.170	0.299	0.185	0.309	0.595	0.611
	Avg	0.125	0.250	0.218	0.323	0.213	0.320	0.182	0.297	0.229	0.344	0.239	0.349	0.227	0.340	0.156	0.278	0.151	0.272	0.145	0.270	0.165	0.286	0.584	0.608
Weather	64	0.001	0.018	0.001	0.020	0.001	0.020	0.001	0.020	0.001	0.020	0.001	0.020	0.001	0.020	0.001	0.021	0.001	0.018	0.001	0.019	-	-	0.002	0.033
	96	0.001	0.021	0.001	0.023	0.001	0.023	0.001	0.023	0.001	0.023	0.001	0.023	0.001	0.023	0.001	0.024	0.001	0.021	0.001	0.022	-	-	0.002	0.033
	128	0.001	0.023	0.001	0.024	0.001	0.025	0.001	0.024	0.001	0.025	0.001	0.025	0.001	0.025	0.001	0.026	0.001	0.023	0.001	0.024	-	-	0.002	0.034
	Avg	0.001	0.021	0.001	0.022	0.001	0.023	0.001	0.022	0.001	0.023	0.001	0.023	0.001	0.023	0.001	0.024	0.001	0.021	0.001	0.022	-	-	0.002	0.033
Traffic	64	0.107	0.174	0.139	0.197	0.102	0.158	0.103	0.158	-	-	-	-	-	-	0.113	0.197	0.216	0.314	0.109	0.181	-	-	4.236	1.671
	96	0.110	0.178	0.155	0.211	0.113	0.168	0.113	0.167	-	-	-	-	-	-	0.118	0.201	0.258	0.344	0.114	0.187	-	-	4.277	1.674
	128	0.114	0.182	0.158	0.216	0.121	0.175	0.119	0.174	-	-	-	-	-	-	0.159	0.267	0.269	0.352	0.117	0.191	-	-	4.159	1.655
	Avg	0.110	0.178	0.151	0.208	0.112	0.167	0.112	0.166	-	-	-	-	-	-	0.130	0.222	0.247	0.337	0.113	0.186	-	-	4.224	1.667
Electricity	64	0.204	0.314	0.289	0.368	0.186	0.294	0.180	0.292	0.289	0.377	0.201	0.309	0.194	0.299	0.245	0.346	0.262	0.366	0.230	0.334	-	-	1.932	1.139
	96	0.224	0.326	0.310	0.385	0.208	0.311	0.200	0.309	0.293	0.380	0.220	0.319	0.213	0.312	0.277	0.368	0.297	0.388	0.265	0.356	-	-	2.055	1.174
	128	0.238	0.335	0.323	0.395	0.225	0.324	0.215	0.320	0.293	0.381	0.234	0.329	0.225	0.323	0.292	0.383	0.217	0.322	0.242	0.339	-	-	2.084	1.181
	Avg	0.222	0.325	0.307	0.383	0.206	0.310	0.198	0.307	0.292	0.379	0.218	0.319	0.211	0.311	0.271	0.366	0.259	0.359	0.246	0.343	-	-	2.024	1.165
1st Count		20		0		5		9		0		0		0		0		4		0		0		0	

A lower mse or mae represents a better prediction. Dash (-) indicates that this dataset was used during model pretraining.

TABLE IV
EXPERIMENTAL RESULTS OF ZERO-SHOT FORECASTING WITH EXOGENOUS VARIABLES

Dataset	Models	UniCastX		$\text{Moirai}_{\text{small}}$		$\text{Moirai}_{\text{base}}$		$\text{Moirai}_{\text{large}}$		GTT_{tiny}		$\text{GTT}_{\text{small}}$		$\text{GTT}_{\text{large}}$	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	64	0.080	0.213	0.084	0.221	0.089	0.226	0.095	0.235	0.097	0.234	0.089	0.229	0.095	0.235
	96	0.100	0.239	0.103	0.246	0.113	0.258	0.112	0.258	0.117	0.261	0.112	0.257	0.112	0.258
	128	0.111	0.254	0.112	0.258	0.120	0.266	0.127	0.274	0.129	0.274	0.125	0.273	0.121	0.270
	Avg	0.097	0.235	0.100	0.242	0.107	0.250	0.111	0.256	0.114	0.256	0.109	0.253	0.109	0.254
ETTh2	64	0.195	0.326	0.225	0.358	0.244	0.370	0.244	0.375	0.226	0.351	0.224	0.355	0.215	0.343
	96	0.237	0.366	0.275	0.398	0.305	0.415	0.290	0.413	0.273	0.392	0.280	0.403	0.248	0.374
	128	0.257	0.387	0.295	0.417	0.326	0.435	0.322	0.437	0.299	0.417	0.315	0.434	0.263	0.392
	Avg	0.230	0.360	0.265	0.391	0.292	0.407	0.285	0.408	0.266	0.387	0.273	0.397	0.242	0.370
ETTm1	64	0.035	0.134	0.045	0.154	0.043	0.148	0.050	0.161	0.044	0.151	0.041	0.142	0.043	0.151
	96	0.045	0.154	0.055	0.172	0.052	0.167	0.067	0.189	0.055	0.173	0.054	0.167	0.053	0.172
	128	0.054	0.170	0.068	0.193	0.062	0.185	0.079	0.211	0.067	0.192	0.066	0.188	0.064	0.189
	Avg	0.045	0.153	0.056	0.173	0.052	0.167	0.065	0.187	0.055	0.172	0.054	0.166	0.053	0.171
ETTm2	64	0.095	0.209	0.218	0.334	0.252	0.361	0.190	0.302	0.132	0.248	0.116	0.235	0.117	0.235
	96	0.118	0.242	0.205	0.328	0.254	0.372	0.217	0.333	0.156	0.280	0.153	0.275	0.142	0.268
	128	0.140	0.268	0.232	0.355	0.278	0.393	0.253	0.366	0.182	0.307	0.184	0.306	0.167	0.296
	Avg	0.118	0.240	0.218	0.339	0.261	0.375	0.220	0.334	0.157	0.278	0.151	0.272	0.142	0.266
Weather	64	0.001	0.018	0.001	0.019	0.001	0.020	0.001	0.020	0.001	0.021	0.001	0.018	0.001	0.018
	96	0.001	0.021	0.001	0.022	0.001	0.023	0.001	0.023	0.001	0.024	0.001	0.021	0.001	0.021
	128	0.001	0.022	0.001	0.023	0.001	0.024	0.001	0.026	0.001	0.026	0.001	0.023	0.001	0.022
	Avg	0.001	0.020	0.001	0.021	0.001	0.022	0.001	0.023	0.001	0.024	0.001	0.021	0.001	0.020
Traffic	64	0.106	0.175	-	-	-	-	-	-	0.112	0.187	0.185	0.286	0.105	0.172
	96	0.110	0.179	-	-	-	-	-	-	0.115	0.190	0.215	0.311	0.109	0.177
	128	0.112	0.182	-	-	-	-	-	-	0.118	0.194	0.220	0.316	0.112	0.181
	Avg	0.109	0.179	-	-	-	-	-	-	0.115	0.190	0.207	0.304	0.109	0.177
Electricity	64	0.181	0.293	0.331	0.415	0.200	0.311	0.196	0.304	0.234	0.342	0.219	0.334	0.188	0.302
	96	0.200	0.305	0.349	0.424	0.222	0.325	0.217	0.317	0.262	0.362	0.250	0.354	0.212	0.317
	128	0.215	0.314	0.346	0.423	0.238	0.337	0.227	0.325	0.279	0.373	0.265	0.363	0.229	0.330
	Avg	0.199	0.304	0.342	0.421	0.220	0.324	0.213	0.315	0.258	0.359	0.245	0.350	0.210	0.316
1st Count		26		0		0		0		0		2		8	

Red values indicates the optimal results and blue values indicates the suboptimal results.

From the results in Table V, it is evident that even in the zero-shot setting, UniCastX achieves competitive forecasting performance compared to supervised forecasting methods. This underscores the capability of our method in general forecasting scenarios. Moreover, fine-tuning UniCastX on the training splits of the benchmark datasets further enhances its performance, achieving the best results in 21 cases, whereas the second-best supervised model only achieved the best results in seven cases. These findings clearly demonstrate the superiority of UniCastX as a foundation model for time series forecasting.

D. Ablation Study

The detailed ablation results on model components and training strategies are shown in Table VI. First, removing the resampling strategy (w/o resampling strategy) means training according to the original data proportions, allowing overrepresented datasets to dominate optimization and bias the learned representations, thereby degrading generalization and forecasting performance. In addition, w/o robust normalization applies conventional normalization without accounting for outliers, which can distort normalization statistics and obscure small-scale temporal variations. Furthermore, w/o exogenous variables and w/o timestamp features exclude exogenous variables and timestamp information, respectively. These inputs provide useful information about external influences and periodic patterns for target forecasting. Finally, w/o attention-sharing uses separate parameters for temporal attention and cross-attention, increasing model complexity and the risk of overfitting. In

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TABLE V
EXPERIMENTAL RESULTS OF IN-DOMAIN FORECASTING EXPERIMENTS

Dataset	Models	UniCastX (ZS)		UniCastX (FT)		TimeXer		iTransformer		DLinear		PatchTST		LightTS		TimesNet		Autoformer		Informer	
	Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	64	0.080	0.213	0.079	0.210	0.086	0.220	0.082	0.221	0.076	0.211	0.087	0.225	0.077	0.216	0.096	0.238	0.122	0.277	0.116	0.285
	96	0.100	0.239	0.095	0.233	0.099	0.243	0.090	0.232	0.092	0.236	0.119	0.262	0.095	0.243	0.103	0.248	0.153	0.317	0.225	0.386
	128	0.111	0.254	0.104	0.246	0.109	0.258	0.097	0.241	0.093	0.236	0.145	0.291	0.106	0.256	0.180	0.339	0.273	0.424	0.229	0.382
	Avg	0.097	0.235	0.093	0.230	0.098	0.240	0.090	0.231	0.087	0.228	0.117	0.259	0.093	0.238	0.126	0.275	0.183	0.339	0.190	0.351
ETTh2	64	0.195	0.326	0.198	0.331	0.282	0.417	0.231	0.372	0.210	0.356	0.218	0.356	0.228	0.372	0.243	0.389	0.346	0.479	0.396	0.516
	96	0.237	0.366	0.243	0.375	0.304	0.433	0.256	0.397	0.242	0.385	0.328	0.448	0.265	0.402	0.373	0.484	0.347	0.486	0.382	0.504
	128	0.257	0.387	0.268	0.399	0.362	0.483	0.257	0.400	0.261	0.404	0.331	0.460	0.293	0.426	0.409	0.515	0.361	0.488	0.493	0.575
	Avg	0.230	0.360	0.236	0.369	0.316	0.444	0.248	0.390	0.238	0.382	0.292	0.421	0.262	0.400	0.342	0.463	0.351	0.484	0.424	0.532
ETTm1	64	0.035	0.134	0.035	0.135	0.036	0.138	0.038	0.140	0.035	0.135	0.037	0.142	0.041	0.152	0.042	0.151	0.165	0.315	0.185	0.348
	96	0.045	0.154	0.044	0.153	0.050	0.167	0.049	0.164	0.044	0.155	0.046	0.158	0.046	0.160	0.053	0.172	0.278	0.385	0.427	0.565
	128	0.054	0.170	0.054	0.173	0.058	0.181	0.059	0.183	0.054	0.172	0.057	0.180	0.056	0.178	0.065	0.196	0.181	0.334	0.287	0.436
	Avg	0.045	0.153	0.044	0.153	0.048	0.162	0.049	0.162	0.044	0.154	0.047	0.160	0.048	0.163	0.053	0.173	0.208	0.345	0.300	0.450
ETTm2	64	0.095	0.209	0.088	0.202	0.106	0.230	0.102	0.227	0.095	0.213	0.097	0.215	0.093	0.213	0.137	0.262	0.261	0.398	0.224	0.370
	96	0.117	0.242	0.113	0.237	0.132	0.265	0.128	0.243	0.118	0.247	0.123	0.255	0.118	0.249	0.172	0.303	0.279	0.414	0.260	0.398
	128	0.140	0.268	0.135	0.265	0.157	0.294	0.147	0.286	0.136	0.269	0.144	0.288	0.136	0.270	0.215	0.342	0.386	0.489	0.229	0.382
	Avg	0.118	0.240	0.112	0.235	0.132	0.263	0.126	0.252	0.116	0.243	0.121	0.253	0.116	0.244	0.175	0.302	0.309	0.434	0.238	0.383
Weather	64	0.001	0.018	0.001	0.017	0.001	0.025	0.001	0.027	0.005	0.062	0.001	0.023	0.005	0.057	0.001	0.027	0.004	0.046	0.002	0.033
	96	0.001	0.021	0.001	0.020	0.001	0.026	0.001	0.028	0.006	0.064	0.001	0.028	0.006	0.067	0.001	0.028	0.007	0.066	0.011	0.072
	128	0.001	0.022	0.001	0.022	0.001	0.026	0.001	0.028	0.006	0.065	0.001	0.027	0.006	0.062	0.001	0.028	0.008	0.069	0.004	0.050
	Avg	0.001	0.020	0.001	0.020	0.001	0.026	0.001	0.028	0.006	0.064	0.001	0.026	0.006	0.062	0.001	0.028	0.006	0.060	0.006	0.052
Traffic	64	0.106	0.175	0.100	0.167	0.130	0.216	0.147	0.245	0.133	0.226	0.138	0.232	0.170	0.273	0.128	0.215	0.216	0.321	0.275	0.365
	96	0.110	0.179	0.104	0.171	0.133	0.219	0.156	0.257	0.136	0.228	0.139	0.226	0.171	0.273	0.129	0.214	0.205	0.312	0.286	0.367
	128	0.112	0.182	0.107	0.174	0.135	0.220	0.168	0.273	0.137	0.229	0.143	0.233	0.172	0.274	0.131	0.215	0.209	0.313	0.298	0.383
	Avg	0.109	0.179	0.104	0.171	0.133	0.218	0.157	0.258	0.135	0.228	0.140	0.230	0.171	0.273	0.129	0.215	0.210	0.315	0.286	0.372
Electricity	64	0.181	0.293	0.167	0.279	0.218	0.334	0.218	0.340	0.179	0.297	0.208	0.324	0.200	0.322	0.286	0.396	0.446	0.503	0.689	0.637
	96	0.200	0.305	0.189	0.294	0.253	0.363	0.239	0.353	0.197	0.309	0.240	0.339	0.222	0.336	0.338	0.430	0.464	0.505	0.639	0.616
	128	0.215	0.314	0.206	0.306	0.258	0.362	0.256	0.365	0.211	0.318	0.251	0.350	0.231	0.341	0.342	0.432	0.525	0.544	0.757	0.692
	Avg	0.199	0.304	0.187	0.293	0.243	0.353	0.238	0.353	0.196	0.308	0.233	0.338	0.218	0.333	0.322	0.419	0.478	0.517	0.695	0.648
1st Count		9		21		0		2		7		0		0		0		0		0	

TABLE VI
ABLATION STUDIES

Configuration	Average MSE
UniCastX	**0.114**
w/o resampling strategy	0.125
w/o robust normalization	0.147
w/o exogenous variables	0.120
w/o timestamp features	0.128
w/o attention-sharing	0.117

Average MSE for 64, 96, and 128 horizons forecasting across all benchmarks with different model components is reported.

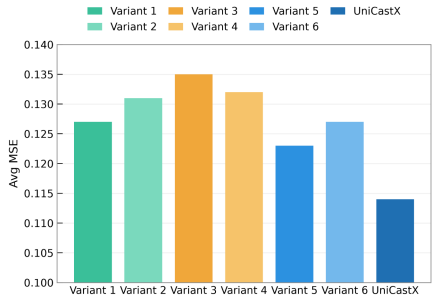


Fig. 3. Comparison between UniCastX and six variants.

contrast, attention sharing improves both generalization and parameter efficiency, making the model more robust and scalable across diverse datasets.

To further investigate the design choices, we construct six variants covering attention, normalization, and data sampling strategies. Variant 1 flattens all target and exogenous patch tokens into a single sequence and models them with global self-attention. Variant 2 first fuses the target and exogenous representations at each patch position and then applies temporal self-attention. Variant 3 replaces the proposed robust normalization with median- and interquartile-range-based scaling, while Variant 4 adopts reversible instance normalization using instancewise statistics. For data sampling, Variant 5 samples all pretraining datasets uniformly, whereas Variant 6 samples domains uniformly and then datasets within each domain according to their relative sizes. As shown in Fig. 3, UniCastX achieves the best overall forecasting performance among these alternatives, providing further empirical support for the adopted designs.

E. Performance Analysis

1) Cold-Start and Small-Sample Adaptation: Due to the varying granularities of time series in real-world scenarios, a universal model should support different context lengths. To evaluate varying input lengths, we test the model on four ETT datasets with a 128-step forecasting horizon, as shown in Fig. 4. We compare models trained with fixed input lengths (dashed lines) and varying input lengths (solid lines). The results show that random masking during training improves the model's ability to handle varying input lengths and enhances generalization. In addition, the model remains robust across different input lengths, even when the input length is shorter than the forecasting horizon, indicating its applicability to limited-history cold-start scenarios.

To further evaluate adaptation under small-sample conditions, we fine-tune UniCastX using 5%, 10%, 25%, and 50% of the training data from four representative datasets. As shown in Fig. 5, forecasting performance progressively improves as more task-specific samples become available. Combined with the competitive zero-shot results reported in Table V, these results show that UniCastX provides effective forecasts without task-specific training and can be further improved with only a limited amount of target-domain data.

2) Robustness on Exogenous Variables: In real-world scenarios, time series data can be affected by intrinsic and extrinsic factors, and exogenous variables with low signal-to-noise ratios may be introduced. To evaluate the robustness of our model, we apply three corruption strategies to the exogenous sequences

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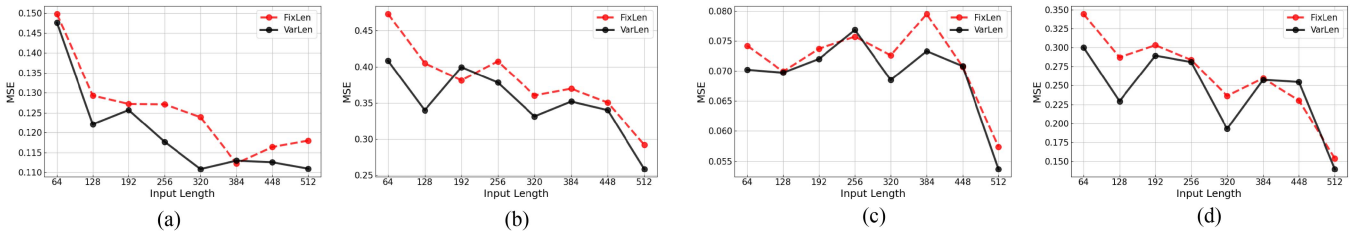


Fig. 4. Comparison of MSE performance between fixed-length and variable-length training models across (a) ETTh1, (b) ETTh2, (c) ETTm1, and (d) ETTm2 datasets for different input lengths.

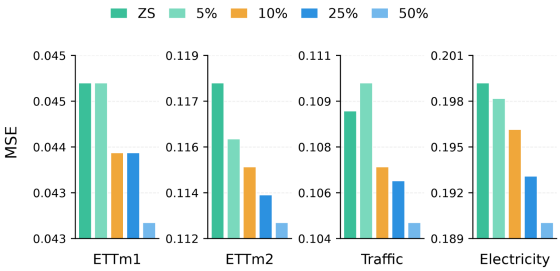


Fig. 5. Small-sample adaptation performance of UniCastX under different proportions of task-specific training data.

TABLE VII
EXPERIMENTAL RESULTS OF MODEL PERFORMANCE UNDER DIFFERENT CORRUPTION STRATEGIES FOR EXOGENOUS VARIABLES

Dataset	ETTh1		ETTh2		ETTm1		ETTm2		Avg	
Strategy	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Noise ($\sigma = 2$)	0.127	0.274	0.304	0.408	0.062	0.185	0.240	0.331	0.183	0.299
Noise ($\sigma = 1$)	0.122	0.268	0.265	0.387	0.061	0.182	0.221	0.312	0.167	0.287
Noise ($\sigma = 0.5$)	0.120	0.269	0.271	0.389	0.060	0.180	0.217	0.303	0.167	0.285
Random	0.121	0.267	0.306	0.417	0.060	0.181	0.234	0.327	0.180	0.298
Zeros	0.121	0.266	0.269	0.394	0.059	0.177	0.148	0.278	0.149	0.279
Ours	**0.111**	**0.254**	**0.257**	**0.387**	**0.054**	**0.170**	**0.140**	**0.268**	**0.140**	**0.270**

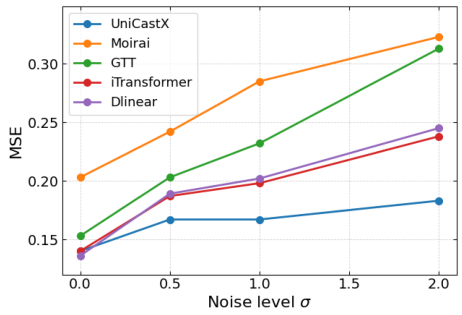


Fig. 6. Robustness comparison under increasing noise levels injected into exogenous variables.

on the ETT datasets, including additive Gaussian noise with varying intensities, random substitution of exogenous values, and removal of exogenous inputs. The results in Table VII show that performance degrades when the exogenous sequences are corrupted. However, even under severe corruption, such as pure noise, our model maintains decent predictive performance. To further compare robustness across different models, we inject increasing levels of noise into exogenous variables at test time.

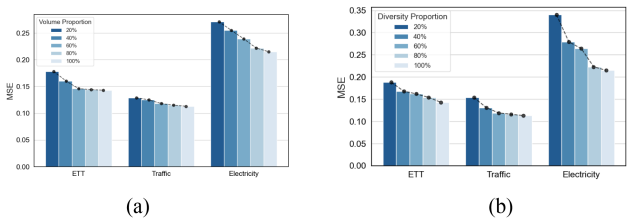


Fig. 7. MSE across the ETT, Traffic, and Electricity benchmarks under (a) varying data volume proportions and (b) data diversity proportions.

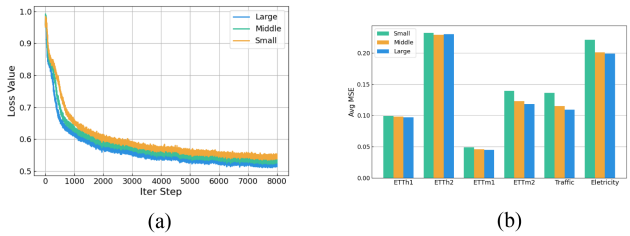


Fig. 8. (a) Training loss curves for models of different scales. (b) Comparison of average MSE across 64, 96, and 128 horizons for models of different scales on benchmarks.

As illustrated in Fig. 6, UniCastX exhibits more graceful performance degradation than representative channel-mixing multivariate baselines as the noise intensity increases, highlighting the robustness of target-centered cross-attention over direct channel mixing. Specifically, C-MSA adaptively weights the aggregation token embeddings of exogenous variables according to their relevance to the target, reducing the influence of corrupted or less informative inputs. Training on diverse time series further improves generalization and robustness to corrupted exogenous data, supporting practical forecasting under high-noise conditions.

3) Effect of Data Volume and Diversity: Fig. 7(a) and (b), respectively, depict the mse across the ETT, Traffic, and Electricity benchmarks under different volumes and diversity of training data. The figures clearly demonstrate that as the volume and diversity of data increase, there is a corresponding improvement in the performance of the model. An increase in data volume enhances the model's ability to capture complex patterns and reduces overfitting by providing more comprehensive temporal coverage, while greater diversity enables the model to generalize better across varying scenarios and handle heterogeneity in

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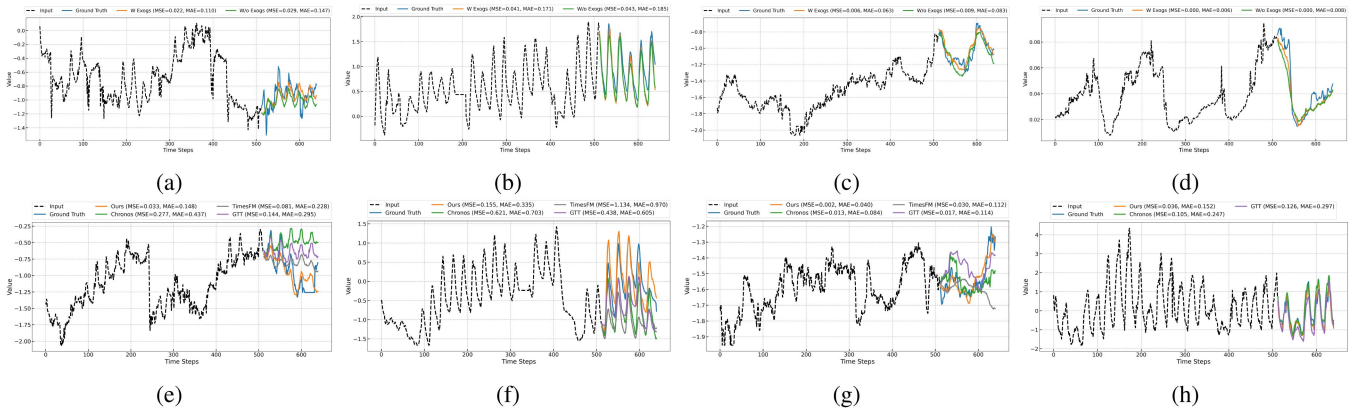


Fig. 9. Visualization results across datasets, with MSE and MAE metrics annotated in all subplots. (a)–(d) Visualization of UniCastX with and without exogenous series. (e)–(h) Comparison of UniCastX with comparison models across datasets.

TABLE VIII
DETAILS OF MODEL ARCHITECTURES WITH DIFFERENT SCALES

Model	Encoder layers	Embedding size	Number of heads	FFN size	Model size
Small	4	384	6	1536	7M
Middle	6	512	8	2048	19M
Large	8	768	12	3072	56M

time series data. In addition, one can find that, compared to increasing data volume, improving data diversity contributes more significantly to model performance. This is because diverse data expose the model to a wider range of temporal patterns, reducing bias and improving adaptability to unseen scenarios. Transformer-based networks are data-intensive and thrive on extensive training data from a wide variety of scenarios [10]. Therefore, it would be intriguing to further explore how the proportions of different, more granular types of time series data impact model performance, with the aim of enhancing it.

4) Scaling Study: To investigate the impact of parameter scale on zero-shot forecasting performance, we trained UniCastX models at three different scales, with structural details provided in Table VIII, where the large version corresponds to the architecture used in our main experiments. Fig. 8(a) illustrates the training loss curves, showing that larger models achieve lower loss values due to their greater capacity, which provides stronger fitting capabilities. Furthermore, as shown in Fig. 8(b), which compares the average mse across various horizons for models of different scales on the benchmarks, it can be observed that zero-shot forecasting performance also improves with larger models. Moreover, considering that the model scale and data volume might not have reached saturation, further increasing the data volume and model scale remains a promising direction for future work.

5) Inference Cost: The inference speed of a forecasting model is a critical factor in real-world applications. In this section, we use the ETT dataset as an example to evaluate the inference efficiency. Specifically, we set the input size to 512, predict the next 128 time steps, and use a batch size of 32 for univariate forecasting experiments. Table IX reports the inference time per batch and compares the performance of

TABLE IX
GPU AND CPU INFERENCE TIME PER BATCH FOR DIFFERENT MODELS

Model	Model size	GPU Time (s)	CPU Time (s)
$\text{Chronos}_{\text{large}}$	710M	21.976	675.314
$\text{Chronos}_{\text{base}}$	200M	8.508	246.874
$\text{Chronos}_{\text{small}}$	46M	2.981	87.482
$\text{Moirai}_{\text{large}}$	311M	5.600	7.390
$\text{Moirai}_{\text{base}}$	91M	3.099	4.172
$\text{Moirai}_{\text{small}}$	14M	2.165	3.478
$\text{GTT}_{\text{large}}$	57M	0.312	1.912
$\text{GTT}_{\text{small}}$	57M	0.244	1.500
GTT_{tiny}	57M	0.176	0.737
TimesFM	200M	0.031	0.220
UniCastX	56M	0.018	0.374

UniCastX with SOTA models. From the experimental results, it can be observed that, despite having fewer parameters, our method consistently outperforms most models in terms of GPU and CPU inference time. Among the compared models, GTT has a similar number of parameters to ours. However, unlike GTT, which employs an autoregressive strategy for next-patch prediction and utilizes cross-attention across patches for all the input series, our method leverages an aggregation token to efficiently extract information from exogenous variables, which enhances computational efficiency. As reported in Table IX, UniCastX achieves an inference latency of 0.018 s per batch on GPU and 0.374 s per batch on CPU, demonstrating its potential for latency-sensitive real-time forecasting applications.

F. Visualization Results

In this section, we present the model's forecasting performance through visualizations for intuitive understanding. The input length of 512 is represented by dashed lines, while the forecast length of 128 is indicated by solid lines in various colors. Fig. 9(a)–(d) illustrates the forecasting performance of our model with and without exogenous variables. From the four cases derived from different datasets, it is evident that incorporating exogenous variables further enhances the forecasting performance of our model. This observation aligns with the experimental results presented earlier, as our model effectively

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extracts useful information from exogenous variables to improve the prediction of target variables. To further showcase the forecasting performance of various foundational models, we conduct a comparative analysis involving the large versions of Chronos, GTT, TimesFM, and our model across six benchmarks. This analysis provides a comprehensive view of each model's capabilities. As shown in Fig. 9(e)–(h), the visual results demonstrate that our model consistently achieves strong performance across these benchmarks, particularly in accurately predicting inflection points and capturing overall trends.

V. Conclusion and Future Work

In this work, we introduce UniCastX, a universal foundation model for zero-shot time series forecasting with exogenous variables, trained on a diverse dataset spanning 26 billion time points across various domains. Unlike existing foundation models primarily designed for univariate or general multivariate forecasting, our approach explicitly distinguishes the forecasting target from auxiliary exogenous variables and selectively incorporates exogenous information to support target forecasting. Compared to SOTA foundation models, UniCastX demonstrates superior zero-shot forecasting capabilities while maintaining low inference time. Moreover, when compared to supervised learning-based approaches, our method exhibits stronger generalization abilities and holds the potential for further fine-tuning to achieve enhanced forecasting performance. In future work, incorporating additional modalities, such as textual descriptions within time series data, to enhance the predictive performance of the model is a promising direction. Furthermore, exploring the universal representations learned by our model for various time series analysis tasks, such as classification, imputation, and anomaly detection, would constitute interesting future work.

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